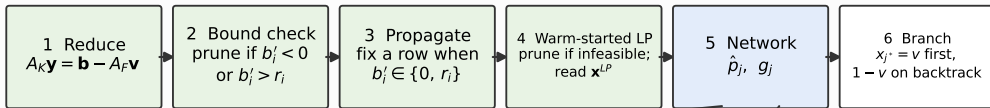


One node of the guided DFS

sound deduction: the only steps that fix variables or prune

learned: chooses, never fixes



Stage 1 (supervised) supplies the first value
 $v = \mathbb{1}[\hat{p}_{j^*} \geq 1/2]$ from the marginal $\hat{p}_j$

Stage 2 (cost-tuned) supplies the branching variable
 $j^* = \arg \max_j g_j, \quad g_j = \log|\hat{p}_j - 1/2| + h_\phi(\mathbf{h}_j)$